

Code Explanation

plot_observations_v0.5 NGC1300 No kinematics.py

This document explains the plotting script that extracts a major-axis light profile from a PHANGS/MUSE FITS image of NGC 1300. The script rotates image coordinates into a disk-aligned frame, selects a narrow slice along the galaxy major axis, plots the image geometry, and saves a binned log-flux profile.

Big Picture

- Load a FITS image from the configured output/data folder.
- Build a rotated coordinate grid centered on the galaxy nucleus.
- Convert pixel coordinates to physical kpc units using the adopted distance and MUSE pixel scale.
- Select pixels in a narrow major-axis slice and keep only positive finite flux values.
- Plot an annotated image map showing the selected slice and bar direction.
- Bin the log flux along the slice and save a radial profile PDF for NGC 1300.

Inputs And Constants

output_folder

The directory where the script expects the FITS file to be located and where it writes the output plots. The current path points to the UCLAN MSc Research/Data folder under the user Dropbox tree.

NGC 1300 parameters

The bottom block hard-codes NGC 1300 geometry: disk PA 278 degrees, inclination 32 degrees, bar PA 99 degrees, distance 19.6 Mpc, image origin (745, 453), and a bar radius derived from 435 MUSE pixels.

Pixel scale

The code assumes MUSE data with 0.2 arcsec per pixel, then converts pixels to kpc using distance * 1000 / 206265.

```
scale_pixel_to_kpc = 0.2 * (distance * 1e3) / 206265.0
slice_width_pixel = slice_width / scale_pixel_to_kpc
```

Helper Function

plot_rotated_coords(data_shape, origin, angle_deg)

Creates x/y pixel index grids, recenters them on the supplied origin, and rotates the coordinate system clockwise by the requested angle. It returns x_rot and y_rot arrays with the same shape as the image.

This lets the rest of the script work in a coordinate frame where the galaxy disk major axis is approximately horizontal. The call inside `image_profile` uses `PA_disc - 90` as the rotation angle.

```
theta = np.deg2rad(-angle_deg)
x_rot = x_shifted * cos(theta) - y_shifted * sin(theta)
y_rot = x_shifted * sin(theta) + y_shifted * cos(theta)
```

Main Routine

`image_profile(...)`

The central routine. It loads the FITS image, extracts a major-axis slice, and optionally creates two figures: an image map JPEG and a flux-profile PDF.

FITS loading

The code opens the FITS file with `astropy.io.fits` and uses extension `[1].data`. It rejects images with no positive finite flux values.

Display scaling

For plotting, it computes `zmin` and `zmax` from the 5th and 99.5th percentiles of positive image flux. Contour levels are geometrically spaced between those limits.

Slice selection

Pixels are selected where `abs(y_rot)` is less than half the requested slice width in pixels. This extracts a strip parallel to the rotated x axis.

Flux profile

The selected flux values are converted to `log10` flux, sorted by x position, trimmed to the requested x limits, and then binned with `scipy.stats.binned_statistic`.

```
ind = abs(y_rot) < slice_width_pixel / 2
valid = isfinite(x_slice) & isfinite(flux_slice) & (flux_slice > 0)
F = np.log10(flux_slice[valid])
```

Image Map Output

When `fig_image` is `True`, the script builds a grayscale `pcolormesh` of the FITS image in rotated coordinates. It overlays contours, shades the selected major-axis slice, and draws a dotted line for the bar direction using the difference between disk PA and bar PA.

- Output filename pattern: `Image_map_<name_out>.jpg`.
- Axes are labelled in kpc by converting rotated pixel ticks with `scale_pixel_to_kpc`.
- The title includes galaxy name, disk inclination, disk-bar PA difference, and slice width.

Profile Plot Output

When `fig_profile` is True, the script creates a profile plot of log flux versus position along the selected slice. It scatters every tenth pixel value, then overlays the median binned profile and a grey 16th-to-84th percentile envelope.

- Output filename pattern: `Obs_image_profile_<name_out>.pdf`.
- The bar radius is shown with dashed vertical lines at $\pm R_{\text{bar}}$.
- A 15 percent bar-radius uncertainty band is shaded around each bar-radius marker.
- The x axis is internally in pixels but displayed in kpc through tick labels.

```
F_bins = stats.binned_statistic(X, F, "median", bins=bins1d)
F_bins16 = binned_statistic(... percentile 16 ...)
F_bins84 = binned_statistic(... percentile 84 ...)
```

Hard-Coded NGC 1300 Run

The executable part of the script changes into `output_folder`, computes the physical bar radius, sets x limits to ± 1.5 bar radii, and calls `image_profile` for the NGC1300 PHANGS image. The script is therefore not a general command-line tool yet; it is a single-target plotting script configured for NGC 1300.

```
Rbar = 435 * (0.2 * (19.6 * 1e3) / 206265.0)
xlims = (-1.5 * Rbar, 1.5 * Rbar)
image_profile(... galaxy="NGC 1300" ...)
```

Notable Assumptions

- The FITS image data are expected in extension [1].
- The image pixel scale is assumed to be 0.2 arcsec per pixel.
- The galaxy center, disk PA, bar PA, distance, and bar radius are supplied manually.
- The code assumes positive flux values are meaningful and excludes zero, negative, NaN, and infinite values from the profile.
- The output folder is hard-coded, and the script changes the process working directory before opening the FITS file.

Summary

In short, this script is an observation-side analogue of a bar-profile extraction step: it rotates a galaxy image into a disk-aligned coordinate frame, slices along the major axis, summarizes flux with robust binned statistics, and marks the expected bar radius on both the image and profile outputs.